★ Data Model v1.0

NAHQ Data Model (V1.0)

1. Users

The **Users** entity represents every authenticated individual who accesses the NAHQ platform. This entity serves as the system's authoritative identity reference and supports authentication, authorization, and audit

ing, and organizational relationships.

A User record is created upon successful onboarding and authentication via an external identity provider (IdP) or integration source.

1.1 Supported User Types

Primary User Categories:

- NAHQ Admin
- Executive User
- Individual Participant

1.2 Data Source

User records may originate from:

- **Nimble (AMS)** – Association Management System
- **Microsoft Entra ID** – For NAHQ administrative users

The system must support multi-source identity federation.

1.3 Attributes

Field Name	Type	Required	Description
user_id	PK (numeric)	Yes	Internal surrogate primary key.
uuid	UUID (unique)	Yes	Public identifier used for APIs and CRUD operations. Prevents exposure of internal numeric IDs.

email	String (unique)	Yes	Primary login identifier. Must be unique across the platform.
name	String	Yes	Full legal name of the user.
role	Enum / Role Reference	Yes	Defines authorization level (Admin, Executive, Participant). Preferably normalized into a Role entity in future iterations.
job_title	String	Optional	Professional title within organization.
years_experience	Integer	Optional	Number of years in healthcare or a relevant field.
organization_id	FK → Organization	Conditional	Associated organization. Required for Executive and Participant users. Nullable for NAHQ Admin (if global).
external_unique_user_id	String	Yes	Identifier from external IdP (e.g., Nimble OAuth ID, Entra Object ID).
external_source	Enum	Yes	Identity provider source (nimble,

			entra_id).
status	Enum	Yes	active, inactive, invited, suspended
created_date	Timestamp	Yes	System record creation date.
updated_date	Timestamp	Yes	System record updation date.
created_by	Timestamp	Yes	User who created the record.
updated_by	Timestamp	Yes	User who last modified the record.
last_login	Timestamp	Optional	Last successful authentication timestamp.
onboarding_completed	Boolean	Yes	Indicates whether the onboarding workflow is fully completed.
profile_completion_percentage	Integer (0–100)	Optional	Tracks profile completeness for UX improvements.

1.4 Rules

1. Email must be unique across the platform.
2. UUID must be globally unique and immutable.
3. A User must be associated with exactly one external identity source.
4. Executive and Individual Participants must belong to an Organization.
5. NAHQ Admin users may optionally belong to an organization but operate at a global scope.
6. The system does not store passwords.

2. Organization

The **Organization** entity represents healthcare institutions participating in the NAHQ system. It acts as the parent entity for Executive Users and Individual Participants.

2.1 Date Source

- Nimble (AMS)

Organization data may initially be synchronized from Nimble and later enriched within NAHQ.

2.2 Attributes

Field Name	Type	Required	Description
org_id	PK (numeric)	Yes	Internal surrogate primary key.
uuid	UUID (unique)	Yes	Public identifier used for APIs and integrations.
name	String	Yes	Legal organization name.
type	Enum	Yes	health_system, hospital, clinic, academic_medical_center, other
parent_org_id	FK → Organization	Optional	self-referential for hierarchy, null if top-level
org_level	ENUM	Yes	ROOT (parent org), SITE, DEPARTMENT
address_line1	String	Yes	Primary address line.
address_line2	String	Optional	Secondary address line.

city	String	Yes	City location.
state	String	Yes	State or region.
postal_code	String	Yes	ZIP or postal code.
country	String	Yes	Country of operation.
size_beds	Integer	Optional	Number of beds (if applicable).
size_employees	Integer	Optional	Number of employees.
primary_contact_id	FK → Users	Yes	Main organizational contact (must be Executive user).
status	Enum	Yes	active, pilot, inactive, suspended
created_date	Timestamp	Yes	Date record was created.
updated_date	Timestamp	Yes	System record updation date.
created_by	Timestamp	Yes	User who created the record.
updated_by	Timestamp	Yes	User who last modified the record.
external_source	Enum	Optional	Source system (nimble).
external_unique_org_id	String	Optional	External AMS identifier.

2.3 Rules

1. Primary Contact must:
   - Belong to the same organization
   - Have an Executive role
2. Suspension of the organization automatically restricts access for associated users.
3. ROOT level must have parent_org_id = NULL.
4. Users must only belong to one ROOT tenant.

Master Data Model

NAHQ Competency Framework, Domain, Role Group & Role Types

The NAHQ Competency Framework is a **version-controlled master data structure** used to:

- Define professional role group & role type classifications
- Map competencies to role types & groups
- Assign scoring weights
- Group competencies into domains
- Support reproducible assessment calculations

This design supports:

- Additions
- Deprecations (no hard deletes)
- Versioned releases
- UI display ordering
- Audit compliance
- Historical assessment accuracy

Because competency definitions and weight values may change(not frequently), the framework is designed using a **snapshot versioning strategy**.

Why Snapshot Versioning?

If weight values or role structures change:

- Historical assessments must NOT be recalculated incorrectly.
- Scores calculated 6 months ago must remain defensible.
- Audit and regulatory reporting must reflect the framework used at that time.

Therefore:

- Each published framework version is immutable.
- New structural changes require a new framework version.
- Only one version is active at a time.
- Historical versions remain archived.

Structural Flow

Framework Version
↓
Organization Standard Type
↓
Role Groups
↓
Role Types
↓
Domains
↓
Competencies
↓
Matrix Weights

All aligned by: framework_version_id

3. Competency Framework Version

The entity represents a complete, immutable release of the NAHQ competency framework. Because weight values, competencies, and role structures may change, the framework is version-controlled using a snapshot-based release model.

At any given time:

- Only one version may be PUBLISHED
- Previous versions are ARCHIVED
- New structural updates are prepared in DRAFT

This ensures historical assessments remain accurate and defensible. This prevents historical recalculation errors and ensures regulatory defensibility.

3.1 Attributes

Field Name	Type	Required	Description
framework_version_id	PK	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier
version_number	Integer	Yes	Sequential version number (1,2,3...)
description	Text	Optional	Summary of changes in this release
status	Enum	Yes	DRAFT, PUBLISHED, ARCHIVED
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	User who created the version
updated_date	Timestamp	Yes	Last update timestamp
updated_by	FK → Users	Yes	The user who last updated the version

3.2 Rules

- Only ONE version can have status = PUBLISHED.
- Archived versions are immutable.
- Publishing a new version automatically archives the previous one.
- All master data tables must reference framework_version_id.

4. Organization Standard Types

Classifies organizations into standardized categories that determine applicable Role Groups.

Each type determines which Role Groups apply.

Examples:

- Large Health System
- Small Hospital
- Community Hospital
- Academic Medical Center

4.1 Attributes

Field Name	Type	Required	Description
org_standard_type_id	PK	Yes	Internal primary key
uuid	UUID	Yes	Public identifier
framework_version_id	FK → Competency_Framework_Version	Yes	Framework reference
name	String	Yes	Standard type name
description	Text	Optional	Detailed explanation
sort_order	Integer	Yes	UI display order
status	Enum	Yes	ACTIVE, INACTIVE
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit

updated_by	FK → Users	Yes	Audit

4.2 Rules

1. Must belong to a framework version.
2. Only ACTIVE types are selectable during onboarding.
3. No physical deletion allowed.

---

4. Domain

The **Domains** entity represents high-level competency categories used for grouping and reporting. Each Domain contains multiple Competencies.

Examples:

- Professional Engagement (PE)
- Quality Leadership & Integration (QLI)
- Performance & Process Improvement (PPI)

4.1 Data Source

NAHQ Framework (static reference data)

4.2 Attributes

Field Name	Type	Required	Description
domain_id	PK	Yes	Internal primary key
uuid	UUID	Yes	Public identifier
framework_version_id	FK → Competency_Framework_Version	Yes	Framework reference
name	String	Yes	Domain name

description	Text	Optional	Detailed explanation
sort_order	Integer	Yes	UI display order
icon	String	Yes	UI icon
color_code	String	Yes	UI color code value
status	Enum	Yes	ACTIVE, INACTIVE
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

4.3 Rules

1. Domain names must be unique within framework version.
2. Domains cannot be deleted if linked to competencies.

---

5. Competencies

The **Competencies** entity defines measurable capability elements. Defines measurable capability elements used in scoring.

Each competency:

- Maps to one or more domains
- Is weighted differently per role
- Is version-controlled

5.1 Data Source

NAHQ Framework (static reference data)

5.2 Attributes

Field Name	Type	Required	Description
domain_id	PK	Yes	Internal primary key
uuid	UUID	Yes	Public identifier
framework_version_id	FK → Competency_Framework_Version	Yes	Framework reference
name	String	Yes	Competency title
description	Text	Optional	Detailed explanation
sort_order	Integer	Yes	UI display order
status	Enum	Yes	ACTIVE, INACTIVE
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

5.3 Rules

1. No physical deletes allowed.
2. Definition changes require new framework version.

---

6. Domain Competency Mapping (Bridge Table)

Bridge table supporting many-to-many relationship:

Domain (1..) ↔ (1..) Competencies. A competency may belong to multiple domains.

6.1 Attributes

Field Name	Type	Required	Description
id	PK	Yes	Primary key
framework_version_id	FK	Yes	Framework version reference
domain_id	FK → Domains	Yes	Domain reference
competency_id	FK → Competencies	Yes	Competency reference
sort_order	Integer	Optional	Display order within the domain
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

6.2 Rules

1. Combination (framework_version_id, domain_id, competency_id) must be unique.
2. Mapping cannot exist across different framework versions.
3. Removing mapping requires a new framework version if already published.
4. Published framework mappings cannot be modified.
5. Domain and Competency must belong to the same framework_version_id.

7. Role Group

The entity represents the high-level functional categorization within the NAHQ competency framework. Each role group contains multiple Role Types.

Examples include:

- Core Quality
- Clinical Quality Bridge
- D&T Clinical Support

Role Groups are version-controlled and may evolve.

7.1 Data Source

NAHQ Framework (static reference data)

7.2 Attributes

Field Name	Type	Required	Description
role_group_id	PK	Yes	Internal surrogate primary key
uuid	UUID	Yes	Public identifier for API exposure
framework_version_id	FK → Competency_Framework_Version	Yes	Links group to specific framework release
name	String	Yes	Name of the role group
description	Text	Optional	Detailed explanation of group scope
sort_order	Integer	Yes	Defines UI display order

status	Enum	Yes	ACTIVE, INACTIVE
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

---

8. Role Type

The entity represents specific professional levels within a Role Group. Each Role Type drives competency mapping and weight calculation.

Examples:

- C-Level/System Executive
- Manager/Supervisor
- Specialist/Analyst

8.1 Data Source

NAHQ Framework (static reference data)

8.2 Attributes

Field Name	Type	Required	Description
role_type_id	PK	Yes	Internal primary key
uuid	UUID	Yes	Public identifier
framework_version_id	FK → Competency_Framework_Version	Yes	Framework reference
name	String	Yes	Role type name

description	Text	Optional	Detailed role definition
sort_order	Integer	Yes	UI order within role group
status	Enum	Yes	ACTIVE, INACTIVE
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

8.2 Rules

1. Reusable across role groups.
2. Version-aligned.

---

9. Organization Standard Type & Role Group Mapping

Defines which Role Groups apply to each Organization Standard Type.

9.1 Attributes

Field Name	Type	Required	Description
id	PK	Yes	Primary key
uuid	UUID	Yes	Public identifier
framework_version_id	FK	Yes	Framework reference
org_standard_type_id	FK	Yes	Organization type

role_group_id	FK	Yes	Role group reference
sort_order	Integer	Optional	UI order
created_date	Timestamp	Yes	Audit
created_by	FK	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK	Yes	Audit

9.2 Rules

1. The combination must be unique within the framework version.
2. Role group and org type must share the same framework version.

---

10. Role Group Role Type Competency Matrix

Defines competency weight values for a specific:

- Role Group
- Role Type
- Competency
- Framework Version

This represents one cell in an Excel sheet. Defines a weighted scoring relationship between a Role Type and Competency within a specific Role Group.

10.1 Attributes

Field Name	Type	Required	Description
id	PK	Yes	Primary key
framework_version_id	FK	Yes	Framework reference

role_group_id	FK → Role_Groups	Yes	Role group
role_type_id	FK → Role_Types	Yes	Role type
competency_id	FK → Competencies	Yes	Competency
weight_value	Decimal	Yes	Scoring weight
created_date	Timestamp	Yes	Audit
created_by	FK	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK	Yes	Audit

10.2 Rules

1. Combination (framework_version_id, role_group_id, role_type_id, competency_id) must be unique.
2. Weight values differ per role group context.
3. Changing weight requires new framework version.

---

11. Audit Logs

The **Audit Logs** entity captures all application-level events across the NAHQ platform.

It records:

- **Who** performed the action
- **What** action occurred
- **When** it occurred
- **Where** it occurred (tenant/system context)
- **Why** (if provided)

Audit logs are system-generated and must not be manually editable.

11.1 Attributes

Field Name	Type	Required	Description
audit_id	PK (numeric)	Yes	Internal surrogate primary key for the audit record.
actor_user_id	FK → Users	Yes	User who performed the action.
actor_user_type	Enum	Yes	Role classification of actor at time of action (NAHQ_Admin, Executive_User, Individual_Participant, System).
action_category	Enum	Yes	High-level classification: ENTITY_ACTION, AUTHENTICATION_EVENT, USER_ACTIVITY, SYSTEM_EVENT.
action_type	Enum	Yes	Specific action performed (CREATE_USER, UPDATE_ORG, LOGIN_SUCCESS, VIEW_DASHBOARD, ACCEPT_INVITE, etc.).

target_type	String	Optional	Logical target of the action. May represent: Entity name (Users, Organization) OR Functional area (Dashboard, Invitation, Authentication).
target_id	String	Optional	Identifier of impacted record (UUID if entity-based). Nullable for general activity events.
actual_reason	String	Optional	Business justification or contextual reason (if provided).
old_value	JSON	Optional	Snapshot of entity state before change (for update/delete events).
new_value	JSON	Optional	Snapshot of entity state after change (for create/update events).
ip_address	String	Optional	IP address from which the action originated.
user_agent	String	Optional	Device/browser/system metadata.

created_date	Timestamp	Yes	Date and time the audit event occurred.

11.2 Rules

1. Audit logs are immutable (no update/delete allowed).
2. All state-changing actions must generate audit records.
3. Authentication attempts (success & failure) must be logged.
4. System events must reference a designated **System User**.

---

12. User Consents

The **User_Consents** entity stores all consent records captured from users within the NAHQ platform. Each consent record represents a **point-in-time agreement** by a specific user to a specific consent version.

12.1 Attributes

Field Name	Type	Required	Description
user_consent_id	PK (numeric)	Yes	Internal surrogate primary key.
uuid	UUID (unique)	Yes	Public identifier used for APIs and external reference.
user_id	FK → Users	Yes	User providing the consent.
consent_type	String / Enum	Yes	Type of consent (TERMS_AND_C

			ONDITIONS, PRIVACY_POLICY, MARKETING_COMMUNICATION, DATA_PROCESSING, etc.).
consent_version	String	Yes	Version identifier of the consent document (e.g., v1.0, 2025-01).
consent_text_snapshot	Text	Optional	Snapshot of the consent text at time of acceptance (recommended for legal defensibility).
consent_status	Enum	Yes	ACCEPTED, DECLINED, WITHDRAWN
consent_method	Enum	Yes	WEB_CLICK, API, SYSTEM_IMPORT, ADMIN_RECORDED
consent_source	String	Optional	Context where consent was captured (Onboarding, Profile Update, Campaign, etc.).
ip_address	String	Optional	IP address at time of consent capture.

user_agent	String	Optional	Device/browser metadata at time of consent.
accepted_date	Timestamp	Yes	Date and time consent action occurred.
withdrawn_date	Timestamp	Optional	Date and time consent was withdrawn (if applicable).
created_date	Timestamp	Yes	Record creation timestamp.
created_by	FK → Users	Yes	Typically same as user_id or System User (if imported).
updated_date	Timestamp	Yes	Last modification timestamp.
updated_by	FK → Users	Yes	User who last updated the record.

12.2 Rules

1. A user may only have one ACTIVE (ACCEPTED) consent per consent_type + version.
2. If a new version is introduced:
   a. Previous version remains historical.
   b. New consent record must be captured.
3. Consent records are never physically deleted (audit requirement).

---

13. Invitations

The **Invitations** entity stores all invitation records across the system. Each record represents a single invitation event issued to a recipient. Stores generic invitation records for onboarding, access control, or participation workflows.

13.1 Attributes

Field Name	Type	Required	Description
invitation_id	PK	Yes	Internal primary key
uuid	UUID (unique)	Yes	Public identifier
invitation_type	Enum	Yes	USER_INVITE, ASSESSMENT_INVITE, etc.
sender_user_id	FK → Users	Yes	User who initiated invitation
recipient_email	String	Yes	Email address of recipient
recipient_user_id	FK → Users	Optional	Linked user if already exists
organization_id	FK → Organization	Optional	Organization context
target_entity_type	String	Optional	Logical target (Assessment, Organization)
target_entity_id	String	Optional	UUID of target entity
invitation_token	String (unique)	Yes	Secure token sent via email
token_expiry_date	Timestamp	Yes	Expiration timestamp

status	Enum	Yes	PENDING, SENT, ACCEPTED, REJECTED, EXPIRED, CANCELLED
accepted_date	Timestamp	Optional	Timestamp of acceptance
rejected_date	Timestamp	Optional	Timestamp of rejection
redirection_url	String	Optional	URL to redirect user after acceptance
metadata	JSON	Optional	Flexible structured data for future extension
created_date	Timestamp	Yes	Audit
created_by	FK → Users	Yes	Audit
updated_date	Timestamp	Yes	Audit
updated_by	FK → Users	Yes	Audit

13.2 Rules

1. The token must expire automatically.
2. Status transitions must be controlled.
3. ACCEPTED invitations cannot be modified.
4. One token = one usage.
5. Expired invitations cannot be accepted.
6. If the recipient already exists, the invitation may link to the existing user.

---

14. Assessment Response

The **Assessment_Responses** entity stores completed or in-progress assessment data collected through Qualtrics and persisted within the NAHQ application.

14.1 Attributes

Field Name	Type	Required	Description
assessment_response_id	PK (numeric)	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier used for APIs
user_id	FK → Users	Yes	User who completed the assessment
organization_id	FK → Organization	Yes	Organization context
framework_version_id	FK → Compentency Framework	Yes	Framework version id
role_group_id	FK → Role_Groups	Yes	Role group context
role_type_id	FK → Role_Types	Yes	Role type selected
qualtrics_survey_id	String	Yes	Qualtrics survey identifier
qualtrics_response_id	String	Yes	Unique response ID from Qualtrics
qualtrics_distribution_id	String	Optional	Distribution identifier (if used)

qualtrics_contact_id	String	Optional	Contact ID in Qualtrics (if synced)
assessment_type	Enum	Yes	BASELINE, FOLLOW_UP, REASSESSMENT, PILOT, etc.
response_json	JSONB	Yes	Raw response payload received from Qualtrics
normalized_response_json	JSONB	Optional	Transformed response mapped to competency IDs
total_score	Decimal	Optional	Calculated total weighted score
completion_time_seconds	Integer	Optional	Time taken to complete survey
started_at	Timestamp	Optional	Assessment start time
completed_at	Timestamp	Optional	Completion timestamp
submission_date	Timestamp	Yes	Date stored in NAHQ system
status	Enum	Yes	STARTED, IN_PROGRESS, COMPLETED, FAILED, CANCELLED
scoring_status	Enum	Yes	PENDING, SCORED, ERROR
scoring_version	String	Optional	Version of scoring algorithm used

is_valid	Boolean	Yes	Indicates whether response passed validation
validation_errors	JSONB	Optional	Validation issues if any
external_reference_id	String	Optional	External tracking reference
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	User or System who created record
updated_date	Timestamp	Yes	Last modification timestamp
updated_by	FK → Users	Yes	User or System who modified record

14.2 Rules

1. response_json must store the original payload unmodified.
2. Scoring must use the matrix aligned to framework_version_id.
3. Each response must reference a single framework_version_id.

---

15. Learning Activity

The **Learning_Activities** entity stores course enrollment, progress, completion, and competency impact data for users participating in learning programs delivered via Oasis LMS.

15.1 Attributes

Field Name	Type	Required	Description
activity_id	PK (numeric)	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier
user_id	FK → Users	Yes	User enrolled in activity
organization_id	FK → Organization	Yes	Organization context
course_id	String	Yes	Course identifier (internal or mapped ID)
course_name	String	Yes	Name of learning activity
course_type	Enum	Optional	COURSE, MODULE, PROGRAM, CERTIFICATION
assessment_id	FK → Assessment_Responses	Optional	Assessment that triggered recommendation
enrollment_date	Timestamp	Yes	Date user enrolled
start_date	Timestamp	Optional	Date course started
completion_date	Timestamp	Optional	Date completed (null if incomplete)
status	Enum	Yes	ENROLLED, IN_PROGRESS,

			COMPLETED, DROPPED, EXPIRED
completion_percentage	Integer (0–100)	Optional	LMS-reported completion percentage
progress_percentage	Integer (0–100)	Optional	Platform-calculated progress percentage
time_spent_hours	Decimal	Optional	LMS-reported time spent
time_invested_hours	Decimal	Optional	Platform-tracked engagement time
quiz_scores	JSONB	Optional	Quiz or assessment scores
credits_earned	Decimal	Optional	CEU/CME credits earned
oasis_activity_id	String	Yes	External Oasis LMS reference ID
oasis_enrollment_id	String	Optional	LMS enrollment reference
competency_impact	JSONB	Optional	Estimated competency improvement impact
expected_completion_date	Timestamp	Optional	Expected completion target date

on_track	Boolean	Optional	Indicates if learner is on track
days_since_enrollment	Integer	Optional	Derived metric
last_synced_at	Timestamp	Yes	Last synchronization with Oasis
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	System or user who created record
updated_date	Timestamp	Yes	Last update timestamp
updated_by	FK → Users	Yes	System or user who modified record

---

16. LMS Course

The **LMS_Courses** entity stores metadata for learning content synchronized from Oasis LMS.

This table represents the master catalog of all available learning activities, independent of user enrollment.

16.1 Attributes

Field Name	Type	Required	Description

course_id	PK (numeric or UUID)	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier for APIs
title	String	Yes	Course title
description	Text	Yes	Detailed course description
competency_mappings	FK → Competencies	Yes	Array of competency IDs addressed by the course
domain_mappings	FK → Domain	Optional	Array of domain IDs supported by the course
duration_hours	Decimal	Yes	Estimated duration in hours
format	Enum	Yes	ONLINE, INSTRUCTOR_LED, HYBRID
difficulty_level	Enum	Yes	FOUNDATIONAL, PROFICIENT, ADVANCED
prerequisites	JSONB	Optional	Array of prerequisite course_ids
provider	Enum	Yes	NAHQ, EXTERNAL
credits	Decimal	Optional	CEU/CME credits awarded
availability_status	Enum	Yes	ACTIVE, ARCHIVED, COMING_SOON

thumbnail_url	String	Optional	Image URL for UI display
oasis_course_id	String	Yes	External reference ID from Oasis LMS
oasis_course_version	String	Optional	LMS version identifier if applicable
external_source	Enum	Yes	OASIS_LMS (future extensibility)
recommendation_weight	Decimal	Optional	Static weighting factor for recommendation engine
last_synced_at	Timestamp	Yes	Last LMS sync timestamp
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	User/System who created record
updated_date	Timestamp	Yes	Last modification timestamp
updated_by	FK → Users	Yes	User/System who modified record

17. AI Conversation

The **AI_Conversations** entity stores all chatbot interactions between users and the NAHQ AI assistant. Stores user–AI chatbot interactions, contextual metadata, audit data, and AI model traceability for compliance and monitoring.

17.1 Attributes

Field Name	Type	Required	Description
conversation_id	PK (numeric)	Yes	Internal primary key
uuid	UUID (unique)	Yes	Public identifier
session_id	String	Yes	Logical conversation thread identifier
user_id	FK → Users	Yes	User interacting with chatbot
organization_id	FK → Organization	Yes	Tenant context
assessment_response_id	FK → Assessment_Responses	Optional	Assessment context referenced
message_type	Enum	Yes	USER_MESSAGE, AI_RESPONSE, SYSTEM_EVENT
user_message	Text	Optional	Raw user input (if USER_MESSAGE)
ai_response	Text	Optional	AI-generated response (if AI_RESPONSE)
context_snapshot	JSONB	Optional	Snapshot of relevant

			scores/content provided to AI
prompt_template_id	String	Optional	Prompt template identifier
confidence_score	Decimal	Optional	AI confidence (if available)
flagged_for_review	Boolean	Yes	Indicates if flagged for admin review
moderation_status	Enum	Optional	SAFE, FLAGGED, REVIEWED
moderation_details	JSONB	Optional	Moderation results metadata
feedback_rating	Integer	Optional	User feedback (1–5)
feedback_comment	Text	Optional	Optional user feedback
conversation_status	Enum	Yes	ACTIVE, CLOSED, ESCALATED
escalation_reason	String	Optional	If escalated to human support
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	Typically system or user
updated_date	Timestamp	Yes	Last modification timestamp
updated_by	FK → Users	Yes	Audit

18. Engagements (Cohort Management)

The **Engagements** entity represents a structured program, cohort, or initiative within an organization. Represents a structured cohort or program engagement within an organization, tracking participants, phases, and performance metrics.

18.1 Attributes

Field Name	Type	Required	Description
engagement_id	PK (numeric)	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier
organization_id	FK → Organization	Yes	Organization context
cohort_name	String	Yes	Cohort name (e.g., “2025 Q1 Cohort”)
wave	Integer	Optional	Wave number (if multiple waves per org)
engagement_type	Enum	Yes	ORGANIZATIONAL, INDIVIDUAL_MEMBER
start_date	Timestamp	Yes	Engagement start date
expected_end_date	Timestamp	Optional	Planned completion date
actual_end_date	Timestamp	Optional	Actual completion date

current_phase	Enum	Yes	INITIATE, ASSESS, PLAN, ACTIVATE
phase_start_dates	JSONB	Yes	Start timestamps per phase
phase_expected_end_dates	JSONB	Optional	Expected end per phase
participant_count	Integer	Optional	Total assigned participants
status	Enum	Yes	PLANNED, ACTIVE, COMPLETE, ON_HOLD, CANCELLED
description	Text	Optional	Engagement notes
program_source	Enum	Optional	PLATFORM, WFA_PROGRAM
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	Creator (Admin)
updated_date	Timestamp	Yes	Last modification timestamp
updated_by	FK → Users	Yes	Last modifier

19. Notifications

The **Notifications** entity stores all outbound and in-app communications generated by the NAHQ platform. Stores system-generated or manual communications sent to users, including audit, delivery tracking, and contextual references.

19.1 Attributes

Field Name	Type	Required	Description
notification_id	PK (numeric)	Yes	Internal surrogate primary key
uuid	UUID (unique)	Yes	Public identifier
organization_id	FK → Organization	Yes	Tenant context
recipient_user_id	FK → Users	Optional	Target user (null if broadcast)
recipient_email	String	Optional	Email address used
recipient_phone	String	Optional	Phone number (if SMS)
sender_user_id	FK → Users	Optional	User initiating communication
notification_type	Enum	Yes	ASSESSMENT_REMINDER, LMS_REMINDER, INVITATION, ENGAGEMENT_ALERT, SYSTEM, REPORT, etc.
communication_channel	Enum	Yes	EMAIL, SMS, IN_APP, PUSH, SYSTEM_EVENT
subject	String	Optional	Subject line (email)

message_body	Text	Yes	Final rendered content sent
template_id	String	Optional	Template reference used
template_version	String	Optional	Template version
related_entity_type	String	Optional	Logical reference (Assessment, Engagement, Learning_Activity, Invitation, etc.)
related_entity_id	String	Optional	UUID of referenced entity
priority_level	Enum	Optional	LOW, NORMAL, HIGH, CRITICAL
scheduled_at	Timestamp	Optional	Scheduled send time
sent_at	Timestamp	Optional	Actual send time
delivery_status	Enum	Yes	PENDING, SENT, DELIVERED, FAILED, BOUNCED
failure_reason	String	Optional	Failure details if applicable
read_status	Enum	Optional	UNREAD, READ
read_at	Timestamp	Optional	Timestamp when read (in-app only)
retry_count	Integer	Optional	Number of retry attempts
external_provider	Enum	Optional	SENDGRID, TWILIO, INTERNAL, etc.

external_message_id	String	Optional	Provider message reference
metadata	JSONB	Optional	Additional structured metadata
is_system_generated	Boolean	Yes	True if auto-triggered
created_date	Timestamp	Yes	Record creation timestamp
created_by	FK → Users	Yes	User/System who created record
updated_date	Timestamp	Yes	Last modification timestamp
updated_by	FK → Users	Yes	Audit